7.3 Higher, faster, stronger (HFS)

In this topic, students use video clips, ICT and laboratory practical activities to study the physics behind a variety of sports:

- speed and acceleration in sprinting and jogging
- work and power in weightlifting
- forces and equilibrium in rock climbing
- forces and projectiles in tennis
- force and energy in bungee jumping.

There are opportunities for students to collect and analyse data using a variety of methods, and to communicate their knowledge and understanding using appropriate terminology.

Students will be assessed on their ability to:	Suggested experiments
1 use the equations for uniformly accelerated motion in one dimension: $v = u + at$ $s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$	
2 demonstrate an understanding of how ICT can be used to collect data for, and display, displacement/time and velocity/time graphs for uniformly accelerated motion and compare this with traditional methods in terms of reliability and validity of data	Determine speed and acceleration, for example, use light gates
3 identify and use the physical quantities derived from the slopes and areas of displacement/time and velocity/time graphs, including cases of non-uniform acceleration	
4 investigate, using primary data, recognise and make use of the independence of vertical and horizontal motion of a projectile moving freely under gravity	Strobe photography or video camera to analyse motion

Students will be assessed on their ability to:	Suggested experiments
6 resolve a vector into two components at right angles to each other by drawing and by calculation	
7 combine two coplanar vectors at any angle to each other by drawing, and at right angles to each other by calculation	
8 draw and interpret free-body force diagrams to represent forces on a particle or on an extended but rigid body, using the concept of *centre of gravity* of an extended body	Find the centre of gravity of an irregular rod
9 investigate, by collecting primary data, and use $\Sigma F = ma$ in situations where m is constant (Newton's first law of motion (a = 0) and second law of motion)	Use an air track to investigate factors affecting acceleration
10 use the expressions for gravitational field strength $g = F/m$ and weight $W = mg$	Measure g using, for example, light gates Estimate, and then measure, the weight of familiar objects
11 identify pairs of forces constituting an interaction between two bodies (Newton's third law of motion)	
12 use the relationship $E_k = ½ mv^2$ for the kinetic energy of a body	
13 use the relationship $\Delta E_{grav} = mg\Delta h$ for the gravitational potential energy transferred near the Earth's surface	
14 investigate and apply the principle of conservation of energy including use of work done, gravitational potential energy and kinetic energy	Use, for example, light gates to investigate the speed of a falling object

Students will be assessed on their ability to:	Suggested experiments
15 use the expression for work $\Delta W = F\Delta s$ including calculations when the force is not along the line of motion	
17 investigate and calculate power from the rate at which work is done or energy transferred	Estimate power output of electric motor (see also outcome 53)
16 understand some applications of mechanics, for example, to safety or to sports	